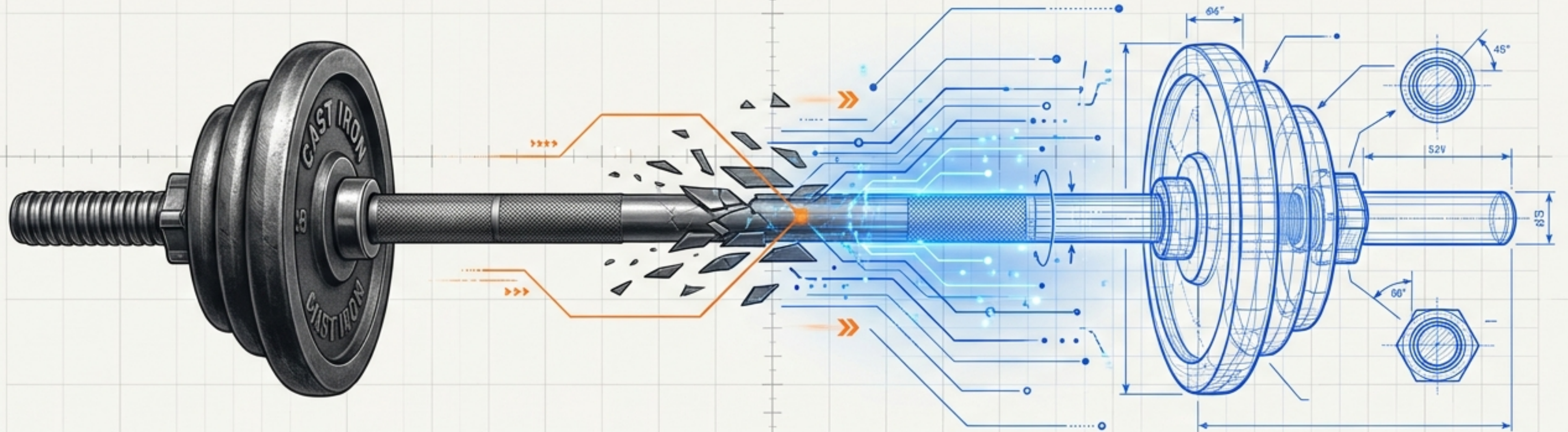


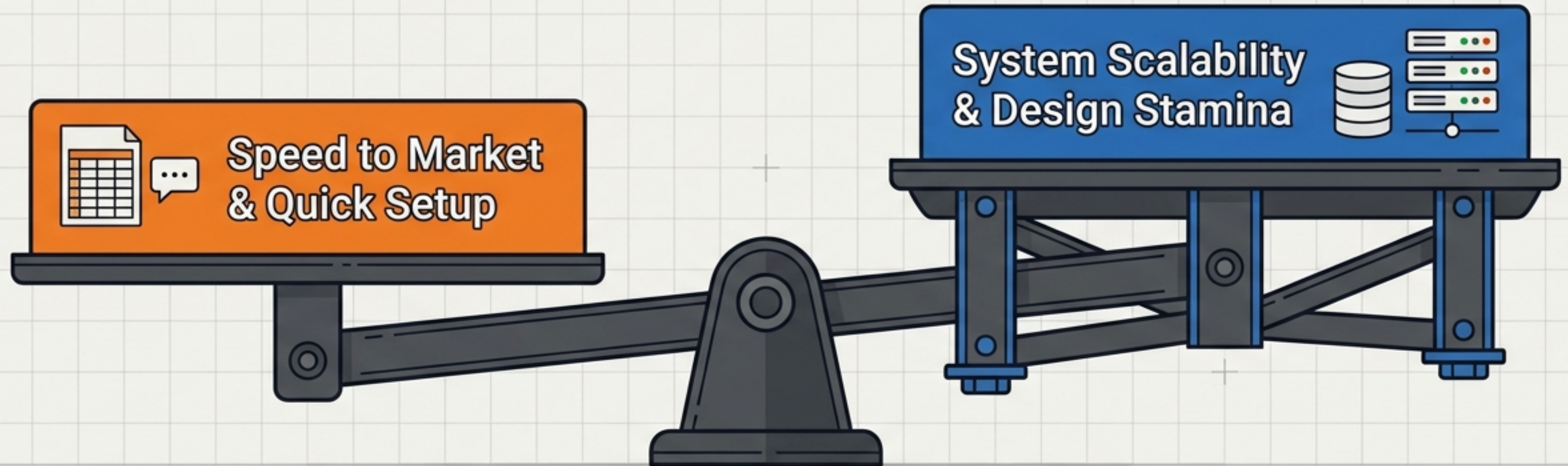
Engineering Design Stamina: From Operational Chaos to AI-Driven Coaching



The Small Goods Gym Technology Roadmap (October – February)

The First Law of Software Architecture

Everything in software architecture is a trade-off.



We are intentionally trading the fragile setup of **scattered spreadsheets** for a **bespoke application infrastructure**, optimizing for massive data loads without system degradation.

The Core Bottleneck: Resolving Systemic Failure

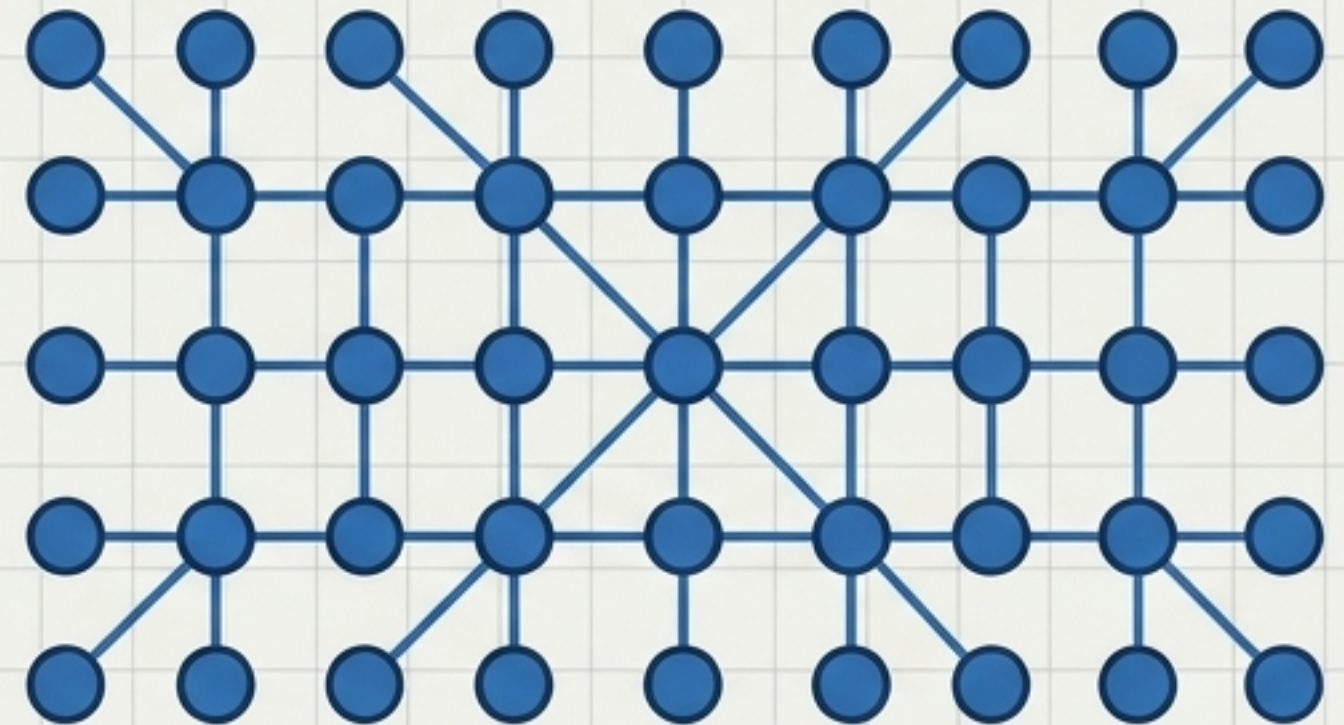
THE SCATTERED STACK

- **Tooling:** Chaotic chat threads & isolated spreadsheets.
- **Data Flow:** Manual, error-prone data entry; isolated silos.
- **Scaling Impact:** High friction; coaches spend hours on admin instead of coaching.

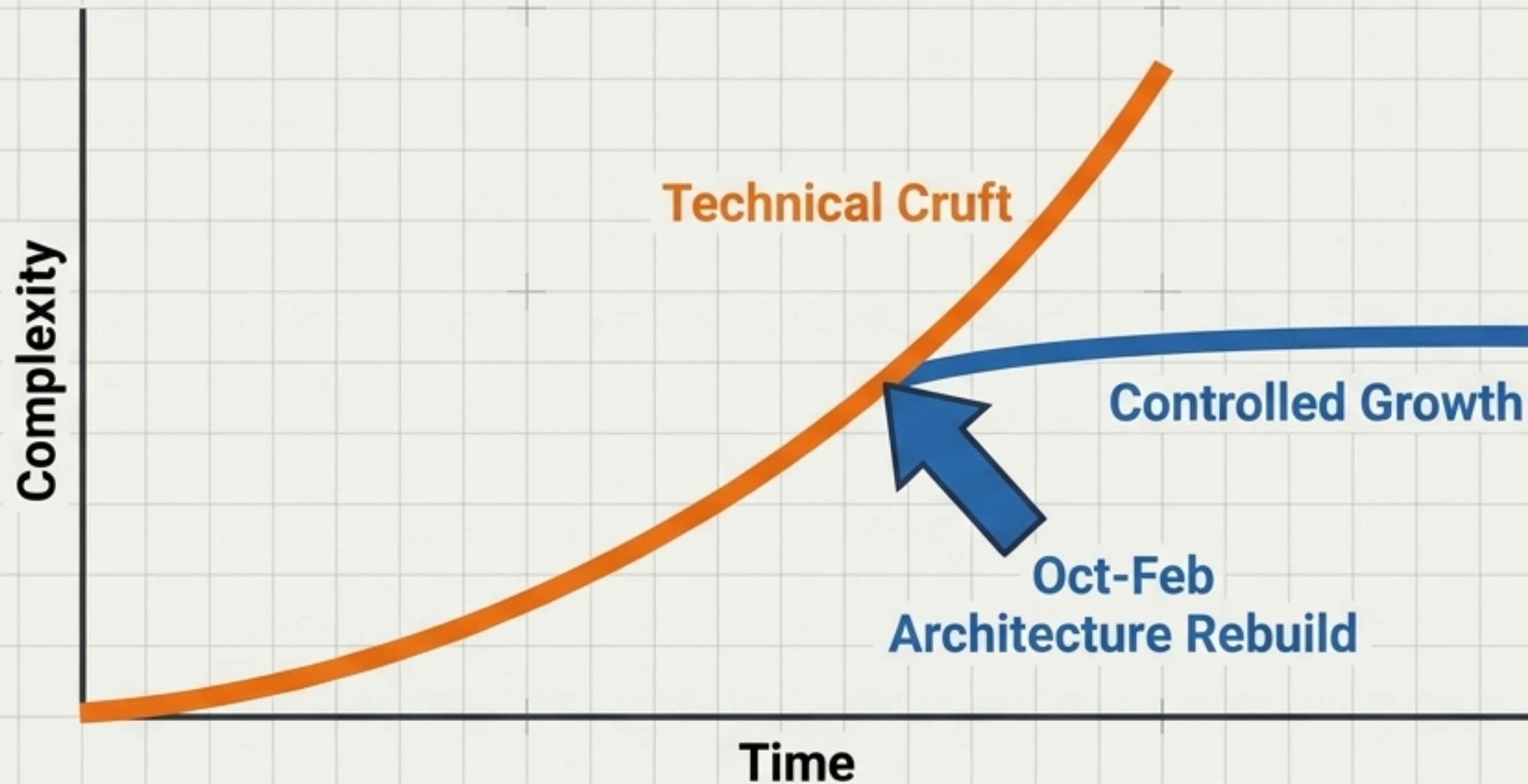


THE UNIFIED ECOSYSTEM

- **Tooling:** Unified application & relational database.
- **Data Flow:** Centralized coaching dashboard; automated synchronization.
- **Scaling Impact:** Frictionless; system absorbs the administrative load.



Managing Complexity: Lehman's Laws of Software Evolution



Law of Increasing Complexity

As software evolves, its complexity naturally increases unless active efforts are made to reduce it.

Law of Declining Quality

The quality of a system will appear to decline unless it is rigorously maintained.

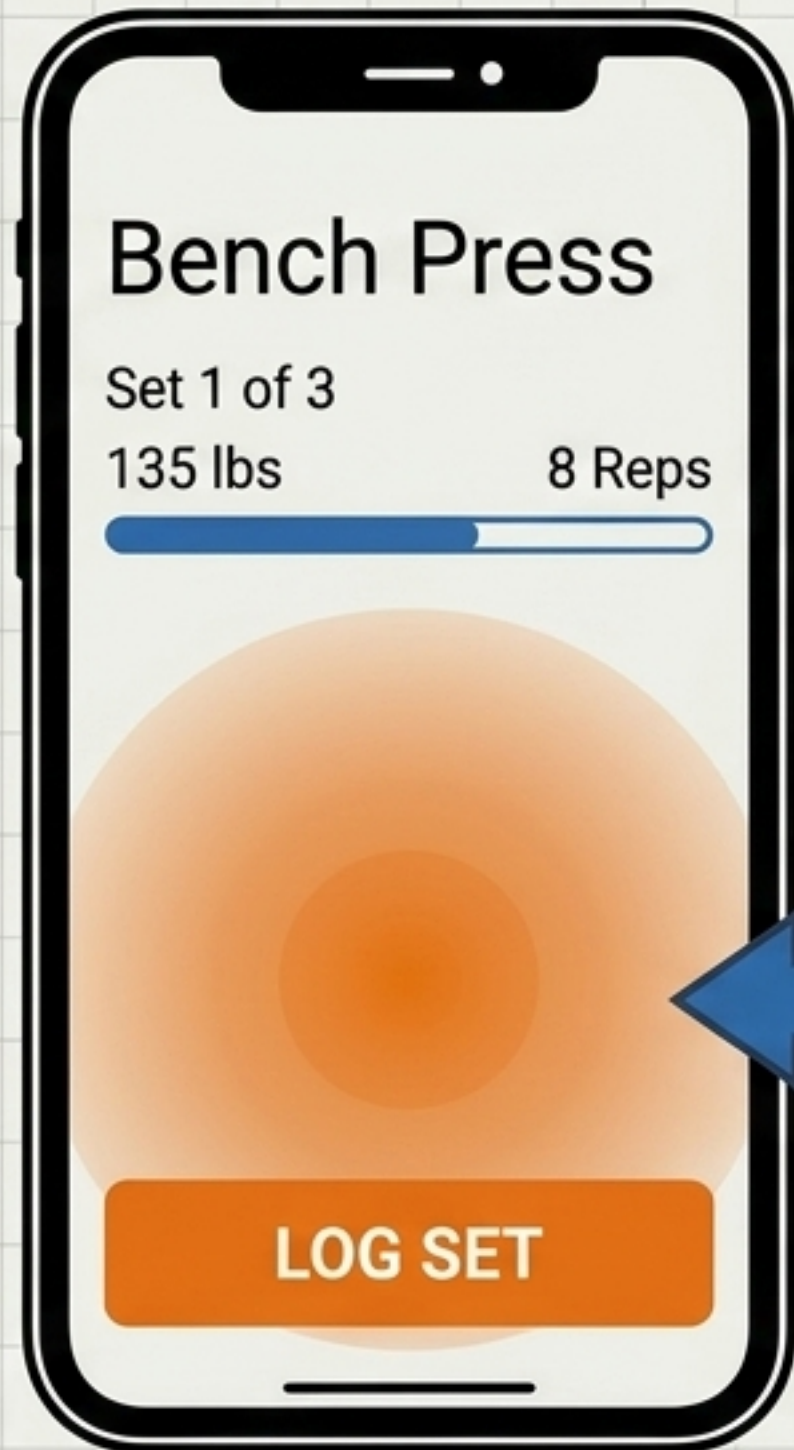
This roadmap is our **hypertrophy phase**. We are proactively rebuilding the system so it doesn't collapse under the weight of new gym members.

Agile Collaboration: The Manifesto in Practice



We will iterate weekly. The ultimate test of the software's validity happens mid-workout on the gym floor, not in a conference room.

Gym-Floor UX Architecture: Designing for the Sweaty Hand



**Thumb
Reach Zone**

Fitts's Law (Physical Constraints)

The time to acquire a target is a function of the distance to and size of the target.

Solution: Massive, thumb-friendly hit-zones for primary actions.

Hick's Law (Cognitive Constraints)

The time it takes to make a decision increases with the number and complexity of choices.

Solution: Eliminate decision fatigue by displaying only one exercise at a time during an active session.

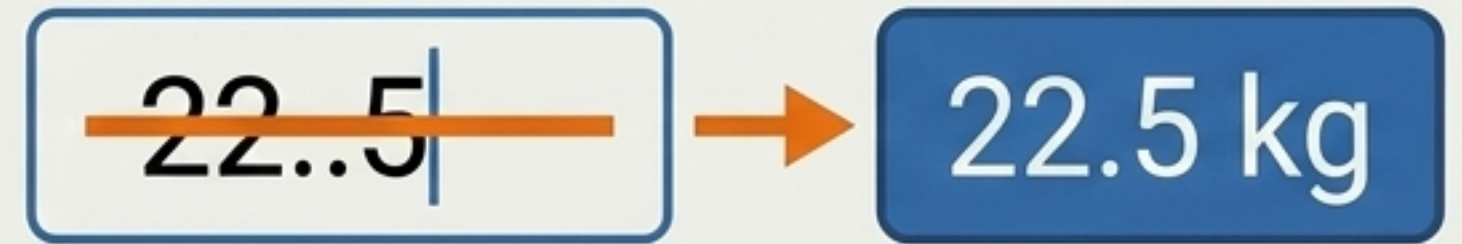
Neurological Flow & Input Resilience

Doherty Threshold



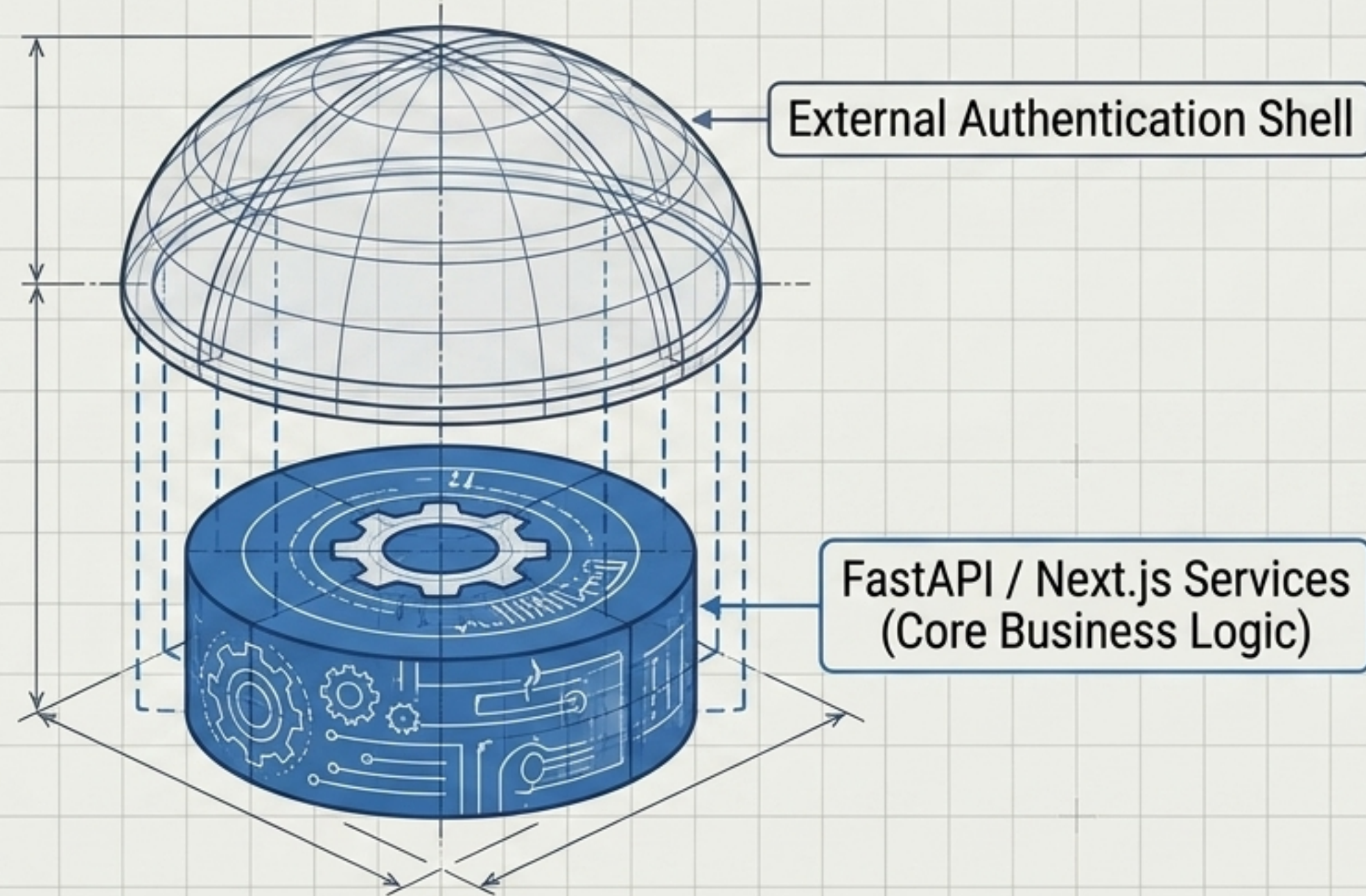
Productivity soars when a computer and its users interact at a pace (<400ms) that ensures neither has to wait. Haptic and visual feedback keeps the lifter in a flow state.

Postel's Law (Robustness Principle)



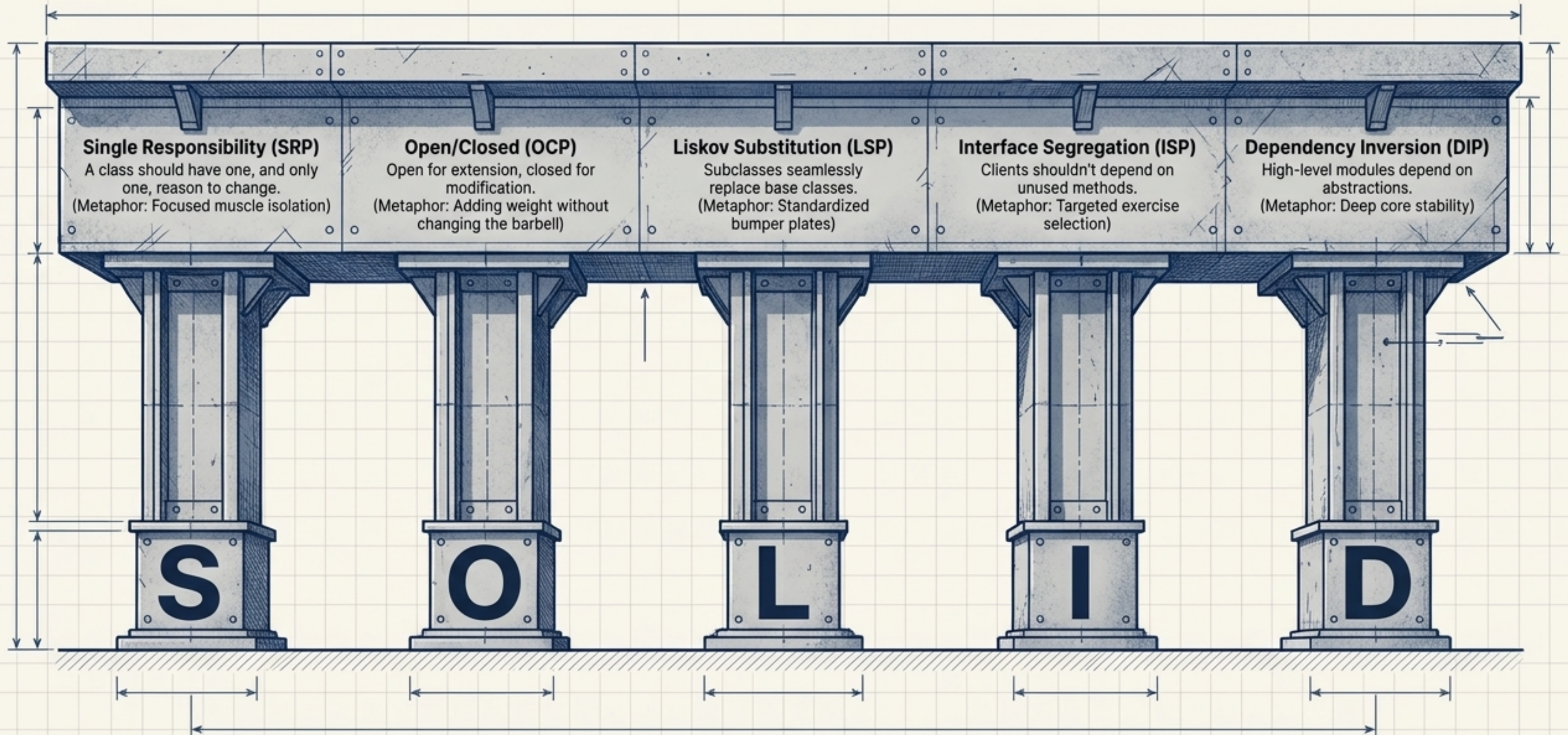
Be conservative in what you do, be liberal in what you accept from others. If a shaking hand mistypes a weight, the app intelligently parses it rather than throwing an error.

Integration & Architecture: Decoupling the Core

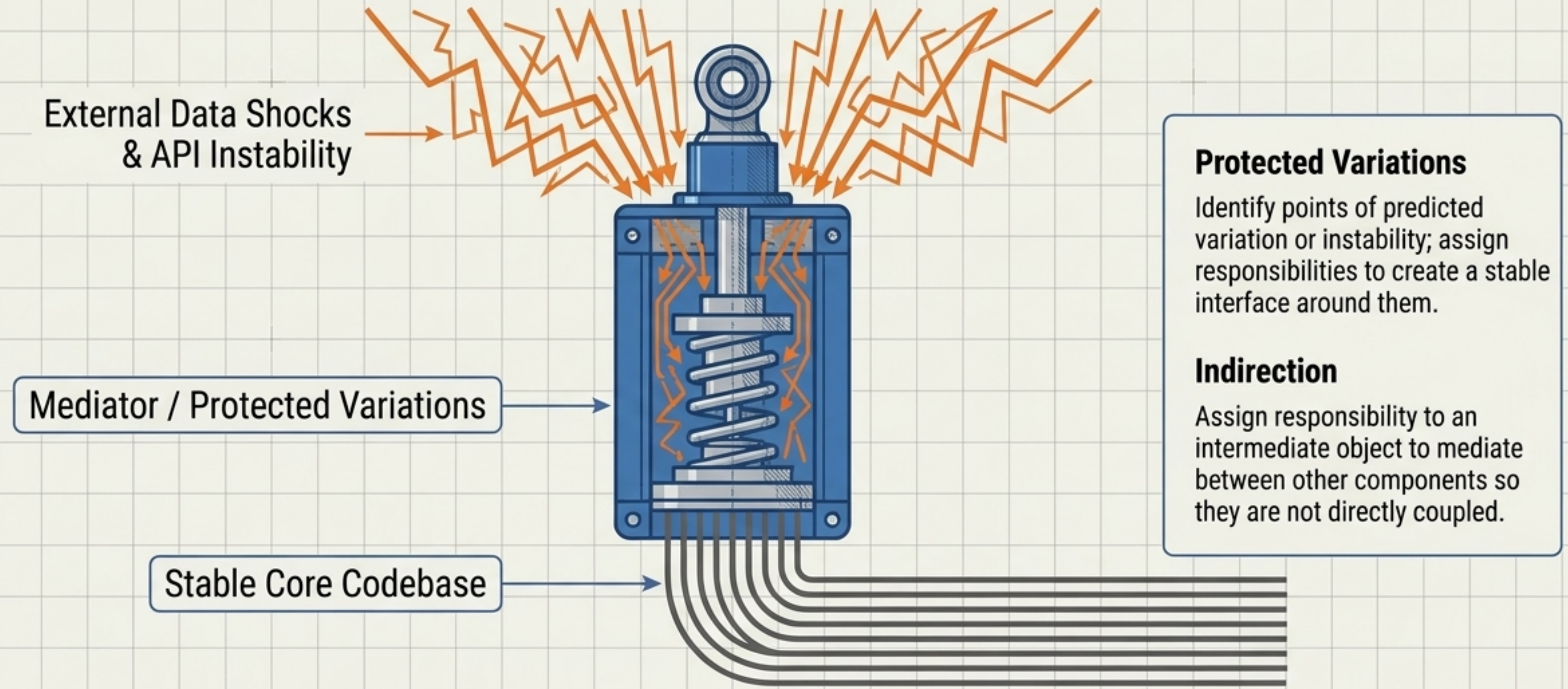


By separating concerns, we protect the vital inner workings. If external dependencies fail, our core API and coaching services remain entirely unaffected.

Building Design Stamina with SOLID Principles

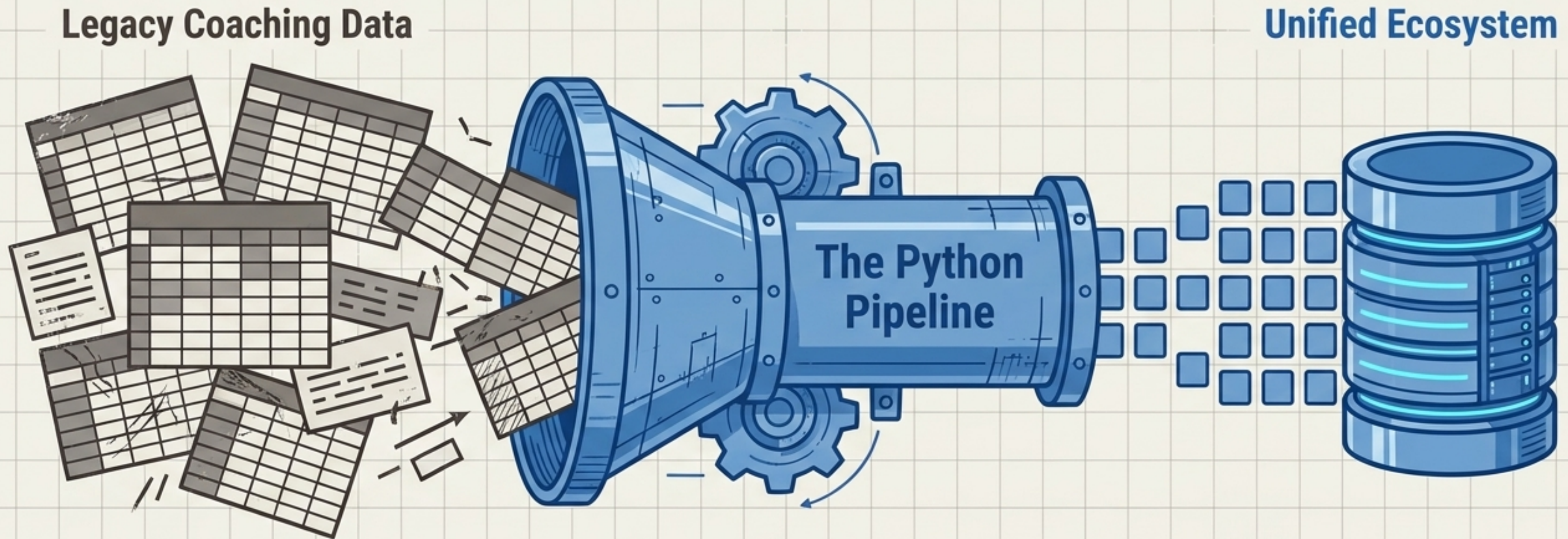


GRASP Patterns: Architectural Shock Absorbers



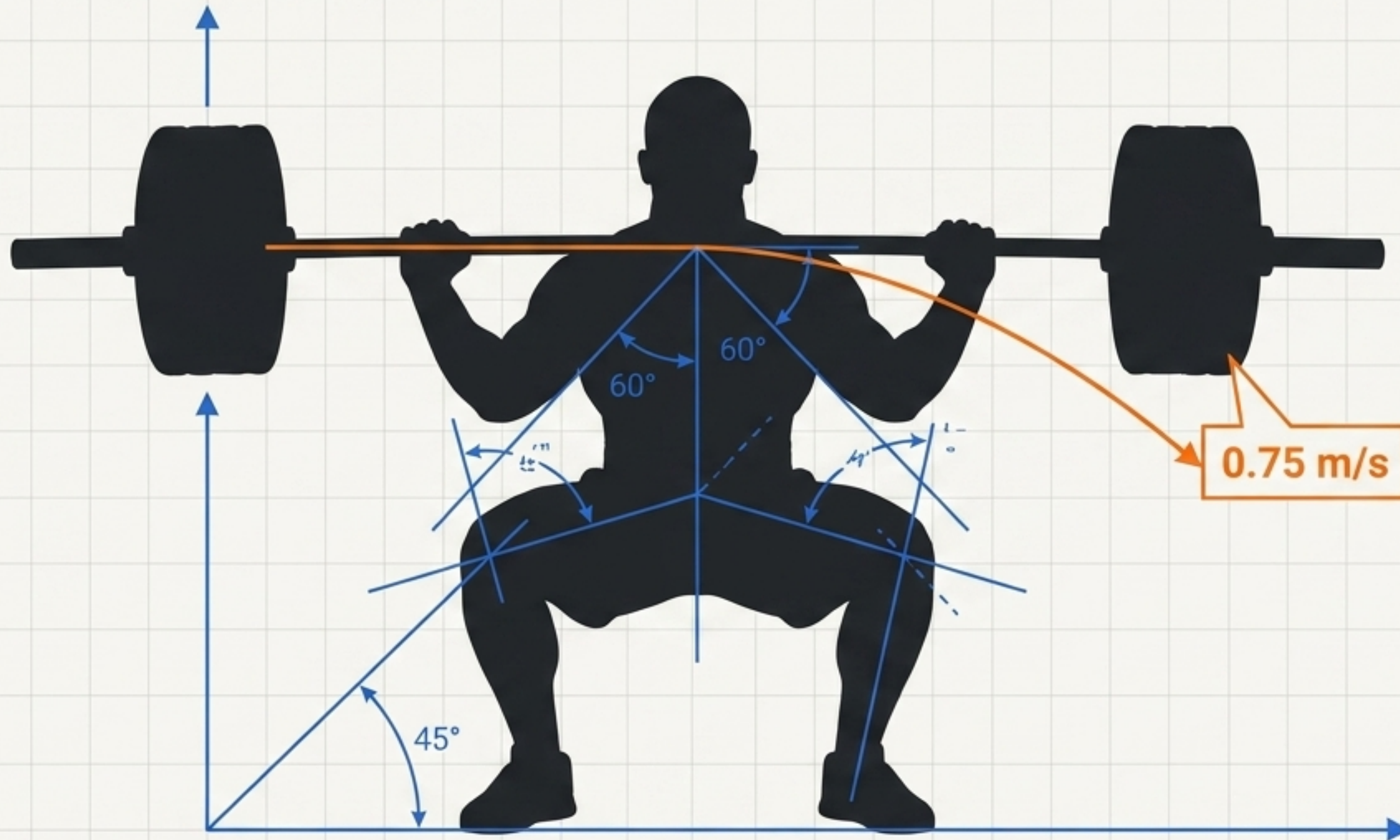
Just as healthy cartilage protects bone from the impact of a heavy squat, these design patterns protect our core codebase from the instability of external tech dependencies.

Seeding the Ecosystem: Python ETL Pipelines



Extract, Transform, Load (ETL). We are standardizing years of legacy data to ensure continuous client progress tracking and to train upcoming AI models.

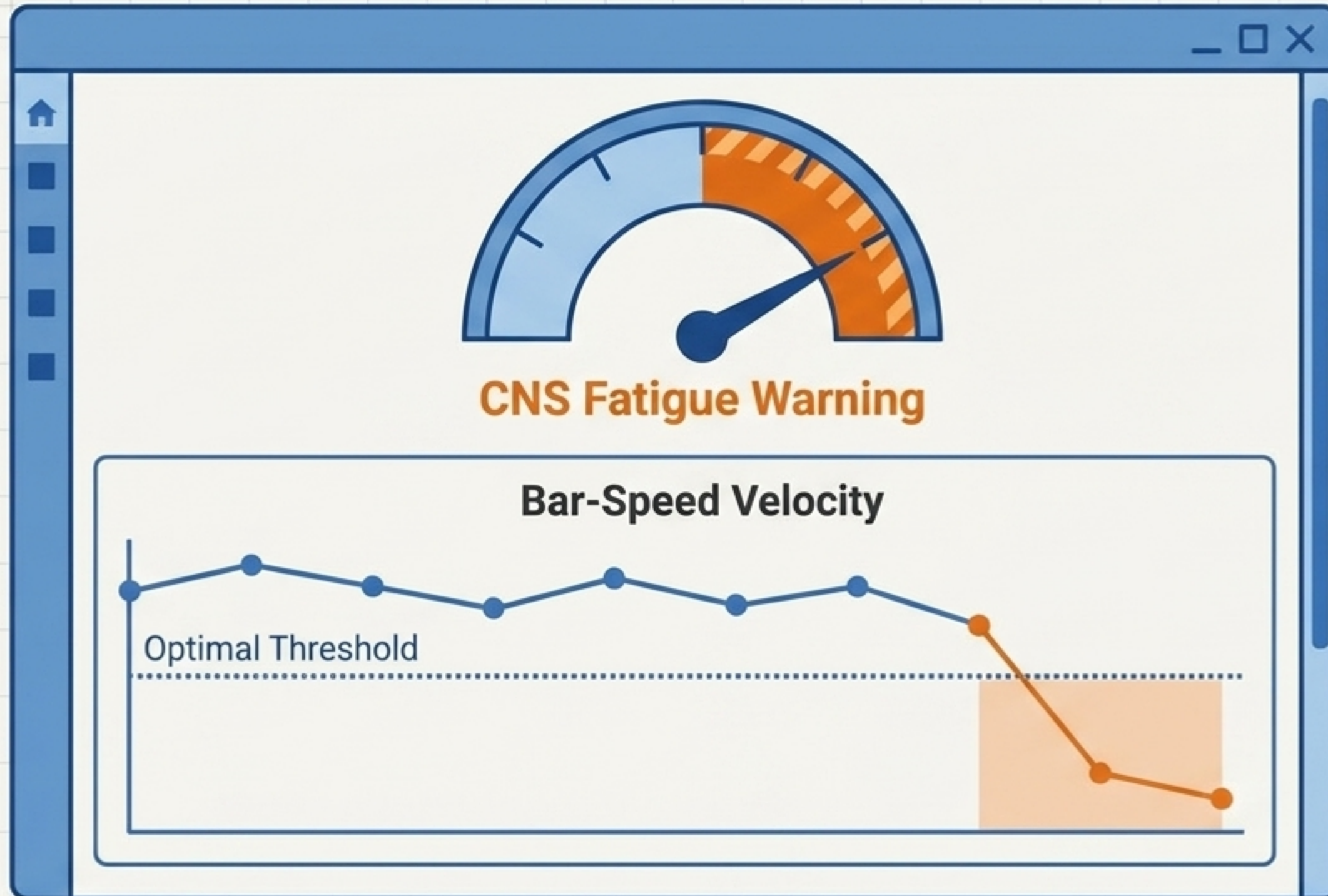
Phase 2 Horizon: The VBT & AI Co-Pilot



The Design Stamina we built allows us to seamlessly ingest advanced hardware data.

We are moving beyond tracking weight and reps to tracking barbell speed (Velocity-Based Training) and precise body mechanics.

Biomechanical Auditing & CNS Fatigue



The Goal

Audit biomechanical Range-of-Motion (ROM) and flag Central Nervous System (CNS) fatigue before an athlete gets injured.

The Ethical Mandate

Aligning with the IEEE-CS/ACM Code of Ethics: Approve software only if there is a well-founded belief that it is safe... and does not diminish quality of life.

The Strategic Delivery Timeline (Oct – Feb)

OCTOBER

Legacy ETL Data Migration & Auth Decoupling

NOVEMBER

FastAPI/Next.js Core Build & SOLID Refactoring

DECEMBER

Gym-Floor UX Beta Testing (Iterative Feedback)

JANUARY

Phase 2 VBT Telemetry & Hardware Integration

FEBRUARY

Full AI Co-Pilot Launch & Biomechanical Auditing

Cultivating Peak Performance



**SMALL
GOODS
GYM**

Honored the First Law of Architecture by choosing scalable design stamina over quick fixes.

Embraced Agile collaboration to ensure the software works in the physical reality of the gym.

Replaced operational chaos with an ecosystem designed to unlock elite, AI-driven coaching.